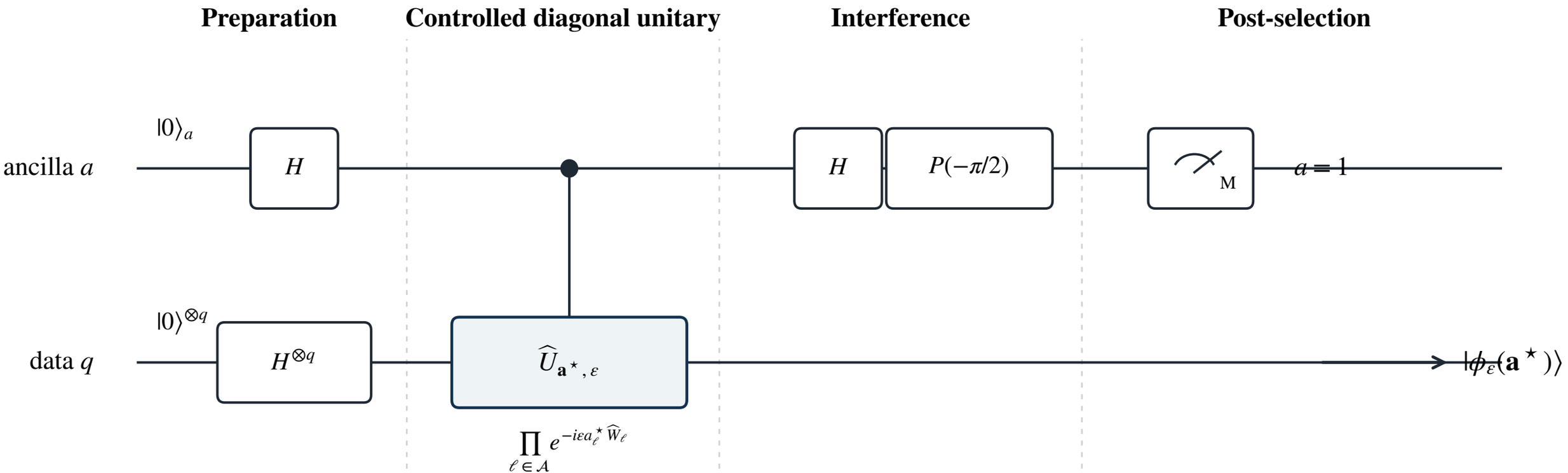

Post-selecting the ancilla isolates $(I - \widehat{U}_{\mathbf{a}^\star,\varepsilon})|+\rangle^{\otimes q}$; for small ε, this approaches the target Walsh-function state.